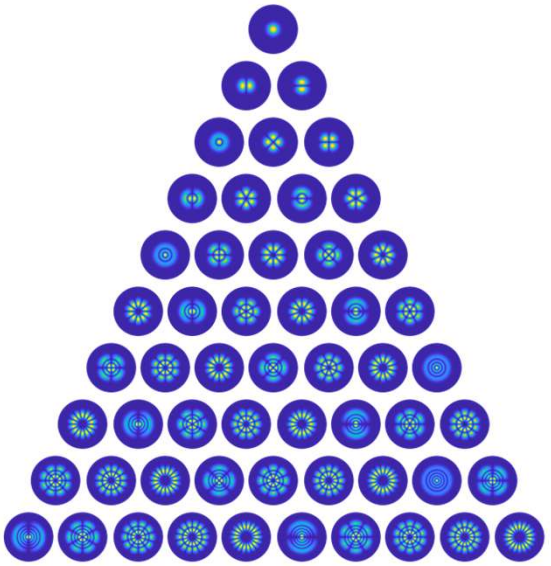
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Advanced optical fiber communications

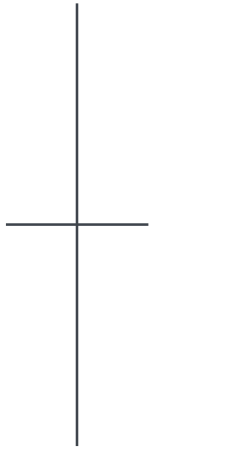
Wintersemester 2025/26

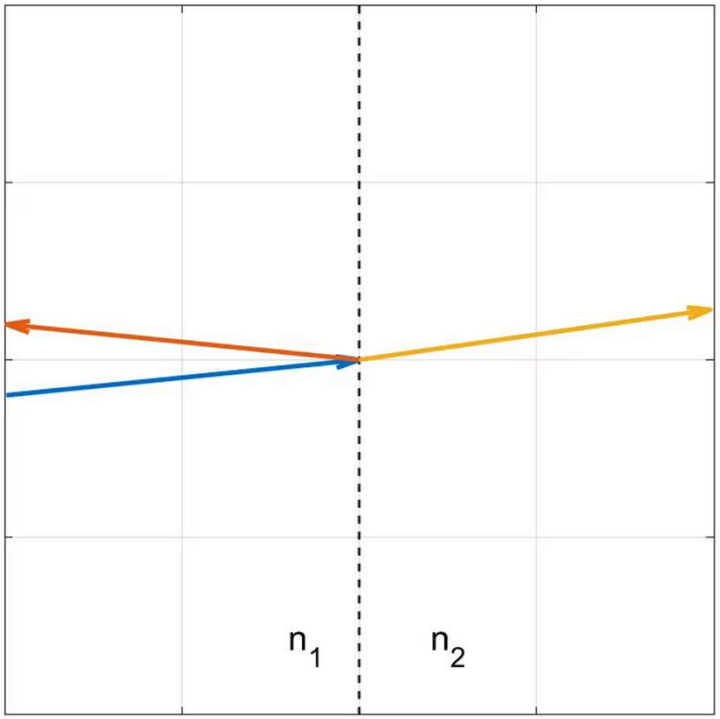
Lecture 4



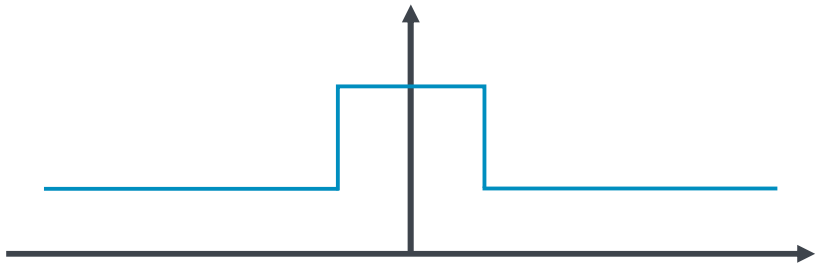
Prof. Dr.-Ing. Georg Rademacher

Snell's law

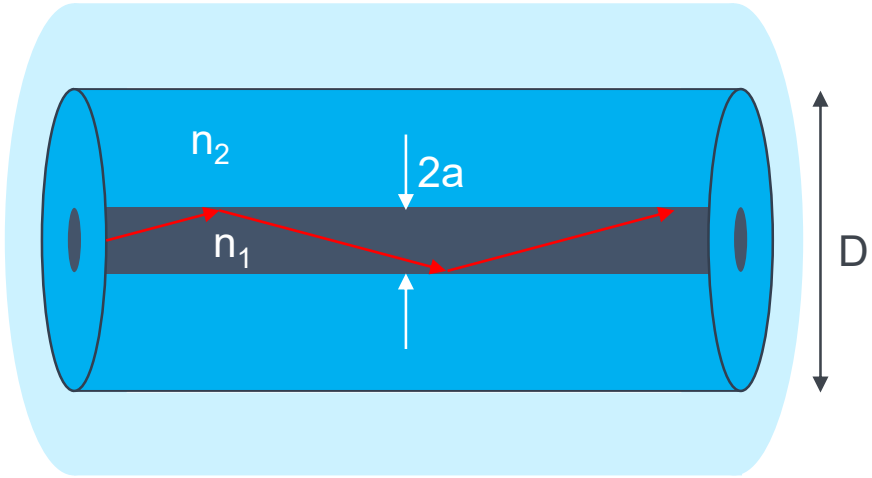




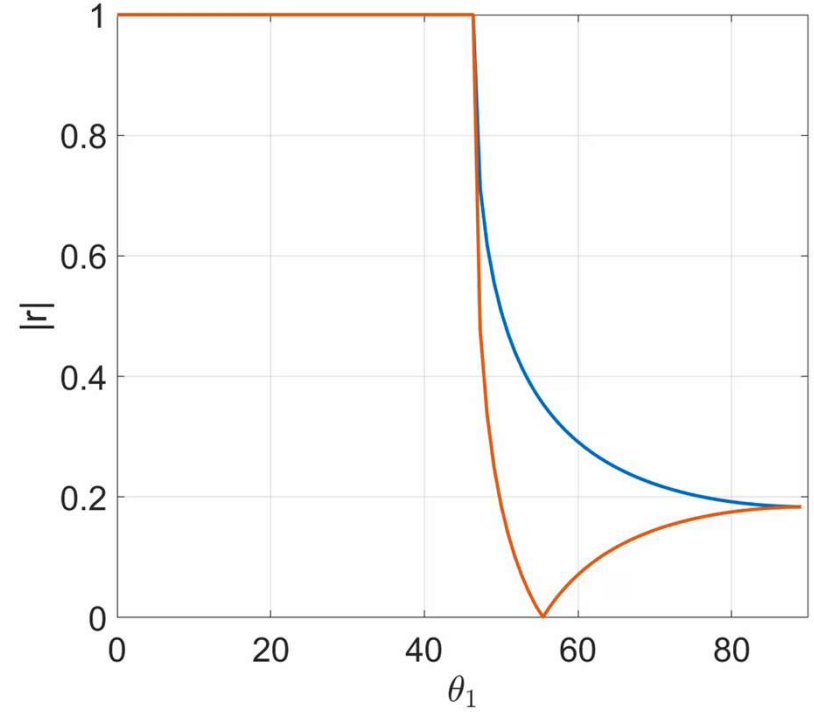
Step-index fiber

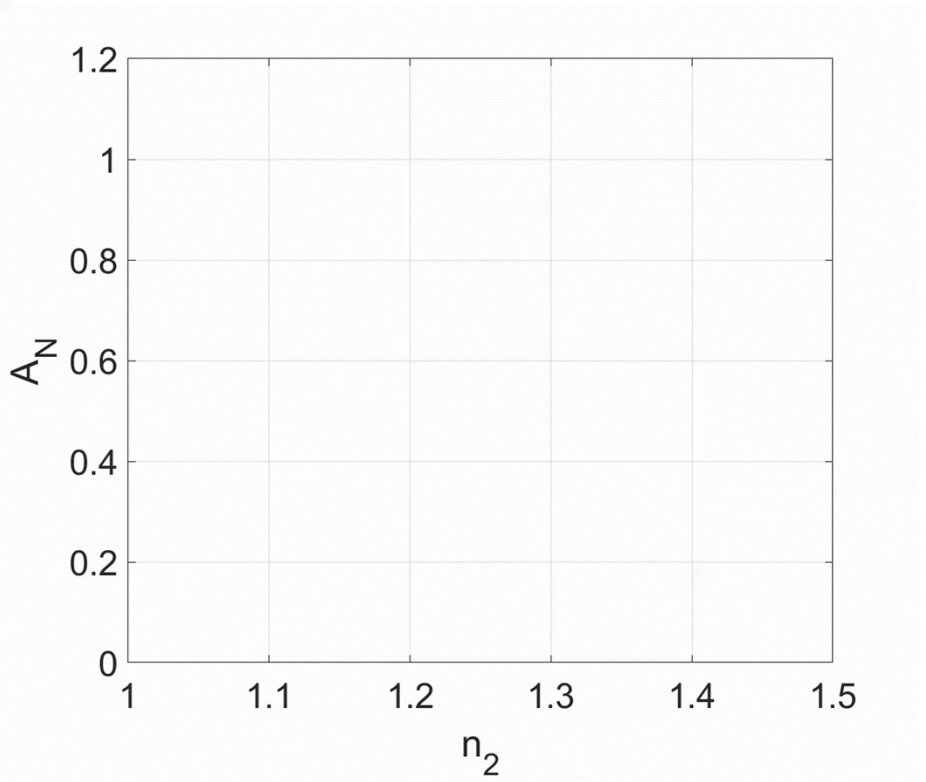


Step-index fiber



Step-index fiber







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Concept Test

Concept test

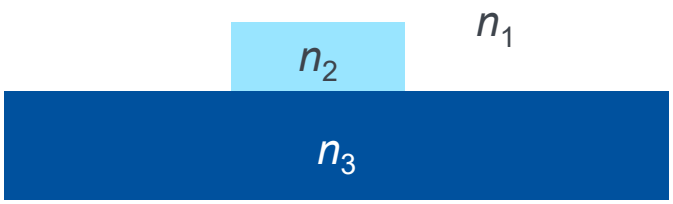
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How are the refractive indices to be chosen for lossless waveguiding in the shown integrated step waveguide?

A $n_3 < n_1 < n_2$

B $n_2 < n_3 < n_1$

C $n_1 < n_2 < n_3$



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Electromagnetc fields in step-index fiber

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$$\underline{\vec{E}}(x,y,z) = \underline{\vec{E}}(x,y) \cdot \exp(-j\beta z)$$

(Eigenwaves)

$$\beta = k_0 \cdot n_1 \cdot \cos(\theta_1)$$

$$0 < \theta_1 < \theta_{1c}$$

$$k_0 \cdot n_2 < \beta < k_0 \cdot n_1$$

Electromagnetc fields in step-index fiber

als auch im Mantel.

$$\vec{E}_{x,y}$$

$$\nabla^2 E_{x,y} + k_0^2 n_i^2 E_{x,y} = 0 \qquad i = 1, 2$$

$$\frac{\partial^2 E_{x,y}}{\partial z^2} = -\beta^2 E_{x,y}$$

$$\nabla_t^2 E_{x,y} + (k_0^2 n_i^2 - \beta^2) E_{x,y} = 0 .$$

$$\nabla_t^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

Electromagnetc fields in step-index fiber

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$$\frac{n_1 - n_2}{n_1} \ll 1 \qquad \text{(weak guidance)}$$

$$\left| k_0^2 n_i^2 - \beta^2 \right| \ll \beta^2$$

$$\beta = k_0 \cdot n_{eff} \qquad n_2 < n_{eff} < n_1$$

Electromagnetc fields in step-index fiber

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$$-\nabla \times \vec{\underline{E}} = j\omega\mu \vec{\underline{H}} \qquad \rightarrow \quad \underline{H}_z = \frac{1}{j\omega\mu_0}\left(\frac{\partial \underline{E}_y}{\partial x} - \frac{\partial \underline{E}_x}{\partial y}\right)$$

$$\nabla \times \vec{\underline{H}} = j\omega\varepsilon_0 n_i^2 \vec{\underline{E}}$$

$$\underline{E}_x = \frac{1}{j\omega\varepsilon_0 n_i^2}\left(\frac{\partial \underline{H}_z}{\partial y} + j\beta \underline{H}_y\right) \qquad \underline{H}_z \approx 0$$

$$\rightarrow \quad \underline{H}_y \approx \frac{\omega\varepsilon_0 n_i^2}{\beta}\underline{E}_x$$

Electromagnetc fields in step-index fiber

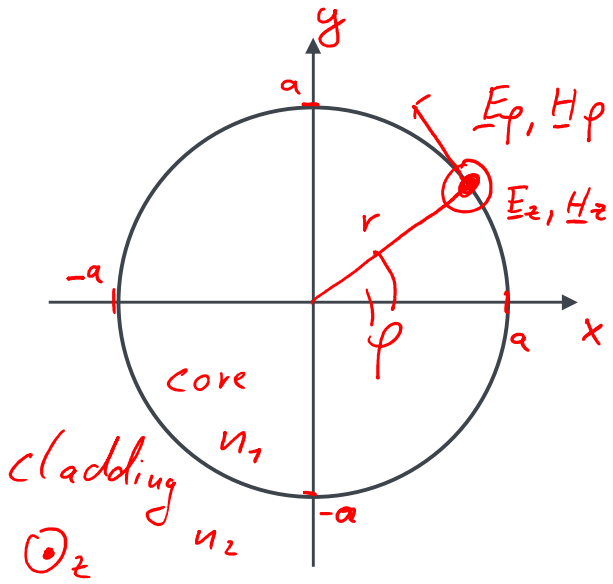
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$$\underline{E}_y = \frac{1}{j\omega\varepsilon_0 n_i^2}\left(-\frac{\partial \underline{H}_z}{\partial x} - j\beta \underline{H}_x\right) \rightarrow \underline{H}_x \approx -\frac{\omega\varepsilon_0 n_i^2}{\beta}\underline{E}_y \qquad \left(\frac{\partial \underline{H}_z}{\partial x} \approx 0\right)$$

$$\underline{E}_z = \frac{1}{j\omega\varepsilon_0 n_i^2}\left(\frac{\partial \underline{H}_y}{\partial x} - \frac{\partial \underline{H}_x}{\partial y}\right) \approx \frac{1}{j\beta}\left(\frac{\partial \underline{E}_x}{\partial x} + \frac{\partial \underline{E}_y}{\partial y}\right)$$

Boundary conditions in step-index fiber

University of Stuttgart
Germany



$r = a \rightarrow E_\varphi, H_\varphi, E_z$ & H_z continous!

$$E_\varphi = E_y \cos(\varphi) - E_x \cdot \sin(\varphi)$$

$$H_\varphi = H_y \cos(\varphi) - H_x \sin(\varphi)$$

$\rightarrow E_x, E_y$: continous @ $r = a$

$\frac{\partial E_x}{\partial \varphi}, \frac{\partial E_y}{\partial \varphi}$: " " "

Boundary conditions in step-index fiber

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$$\underline{E}_z \approx \frac{1}{j\beta}\left(\frac{\partial \underline{E}_x}{\partial x} + \frac{\partial \underline{E}_y}{\partial y}\right)$$

$$\frac{\partial}{\partial x} = \cos(\varphi)\frac{\partial}{\partial r} - \frac{1}{r}\sin(\varphi)\frac{\partial}{\partial \varphi}$$

$$\frac{\partial}{\partial y} = \sin(\varphi)\frac{\partial}{\partial r} + \frac{1}{r}\cos(\varphi)\frac{\partial}{\partial \varphi}$$

$$\underline{E}_z = \frac{1}{j\beta}\left(\cos(\varphi)\frac{\partial \underline{E}_x}{\partial r} - \frac{1}{r}\sin(\varphi)\frac{\partial \underline{E}_x}{\partial \varphi} + \sin(\varphi)\frac{\partial \underline{E}_y}{\partial r} + \frac{1}{r}\cos(\varphi)\frac{\partial \underline{E}_y}{\partial \varphi}\right)$$

$$\rightarrow \quad \underline{E}_x,\ \underline{E}_y,\ \frac{\partial \underline{E}_x}{\partial r},\ \frac{\partial \underline{E}_y}{\partial r} \quad : \text{continuous!}$$



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Concept Test

Concept test

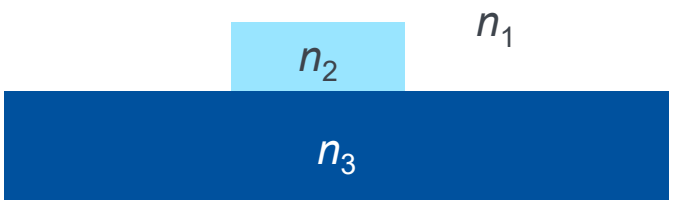
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In which range lies β, if the propagating wave is guided and confined in the shown step waveguide?

A $n_3 < n_1 < n_2 < \beta$ 20%

B $\beta < n_3 < n_1 < n_2$ 20%

C $n_3 < n_1 < \beta < n_2$ 65%



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Linearly polarized (LP) waves

$\underline{E}_x$ & $\underline{E}_y$ are uncoupled !

$\underline{E}_y = 0$ $\qquad$ $\underline{E}_x = \psi(r,\varphi)\exp(-j\beta z)$

$$\psi(r,\varphi) = \begin{cases} \psi_1(r,\varphi), & r \le a \\ \psi_2(r,\varphi), & r > a \end{cases}$$

$$\nabla_t^2 \psi_i + (k_0^2 n_i^2 - \beta^2)\psi_i = 0$$

Linearly polarized (LP) waves

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$$\psi_1\big|_{r=a} = \psi_2\big|_{r=a}$$

$$\frac{\partial \psi_1}{\partial r}\bigg|_{r=a} = \frac{\partial \psi_2}{\partial r}\bigg|_{r=a}$$

normalized frequency

$$V = k_0 \cdot a\sqrt{n_1^2 - n_2^2} = k_0 \cdot a \cdot NA$$

Normalized Prop. constant:

$$B = \frac{\frac{\beta^2}{k_0^2} - n_2^2}{n_1^2 - n_2^2}$$

core parameter $u = V\sqrt{1-B} = a\sqrt{k_0^2 n_1^2 - \beta^2}$

cladding parameter $v = V\sqrt{B}$

Linearly polarized (LP) waves

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$$a^2 \nabla_t^2 \psi_1 + u^2 \psi_1 = 0 \qquad r \le a$$

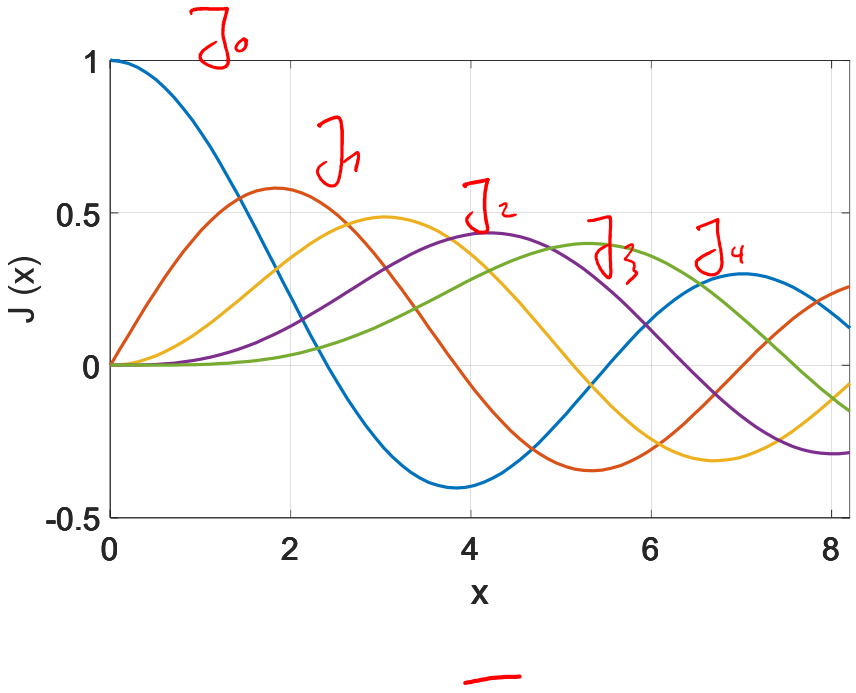
$$a^2 \nabla_t^2 \psi_2 - w^2 \psi_2 = 0 \qquad r \gg a$$

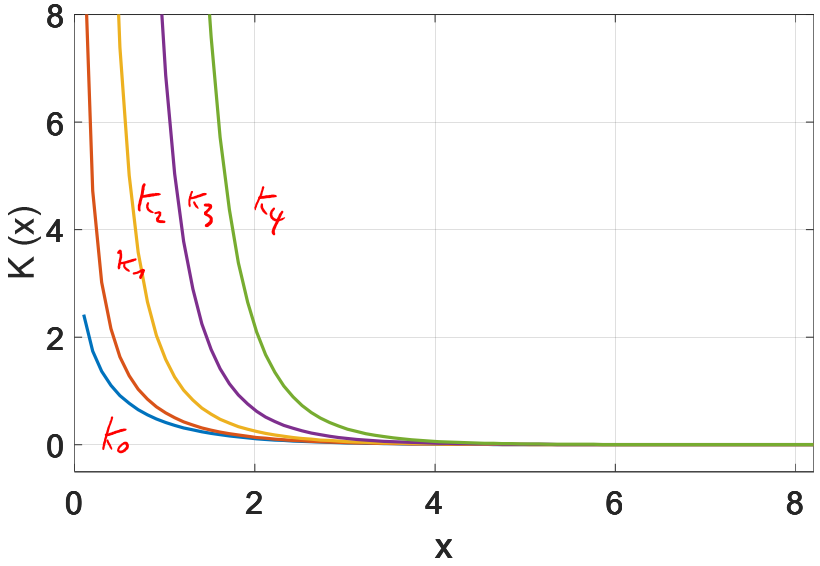
$$\psi_1(r,\varphi) = A_1 J_L\left(r \cdot \frac{u}{a}\right) \begin{Bmatrix} \cos(L\cdot\varphi) \\ \sin(L\cdot\varphi) \end{Bmatrix}$$

$$\psi_2(r,\varphi) = A_2 \cdot K_L\left(r \cdot \frac{w}{a}\right) \begin{Bmatrix} \cos(L\cdot\varphi) \\ \sin(L\cdot\varphi) \end{Bmatrix}$$

Besselfunctions

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Linearly polarized (LP) waves

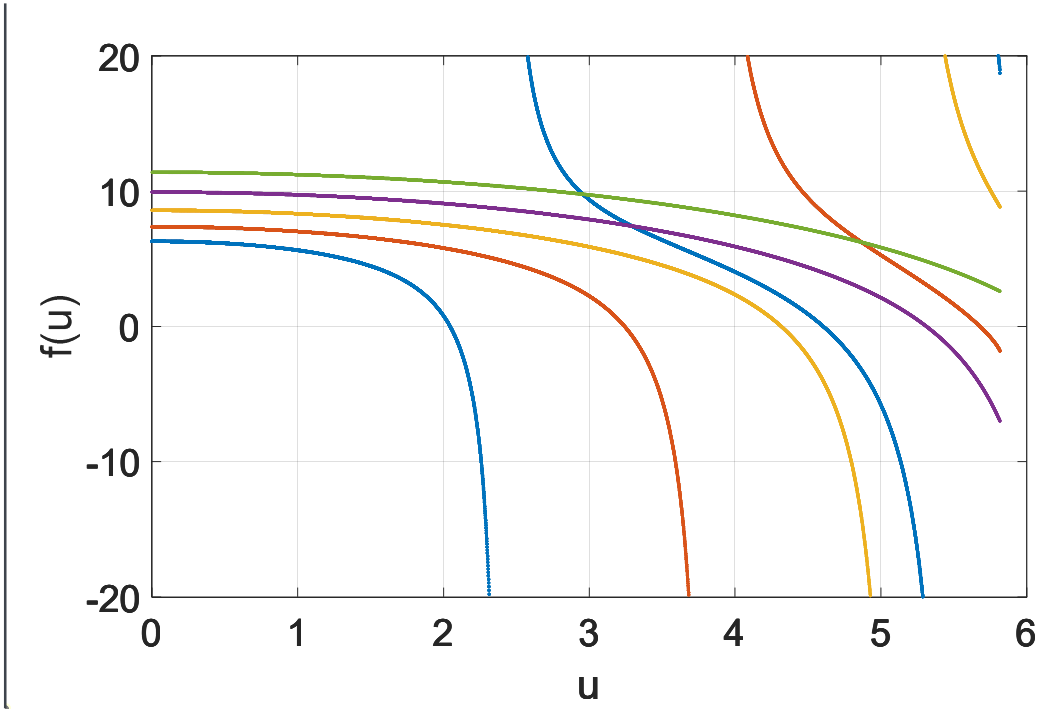
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$$\psi_1\big|_{r=a} = \psi_2\big|_{r=a} \quad \rightarrow \quad A_1 \cdot J_l(u) = A_2 K_l(v)$$

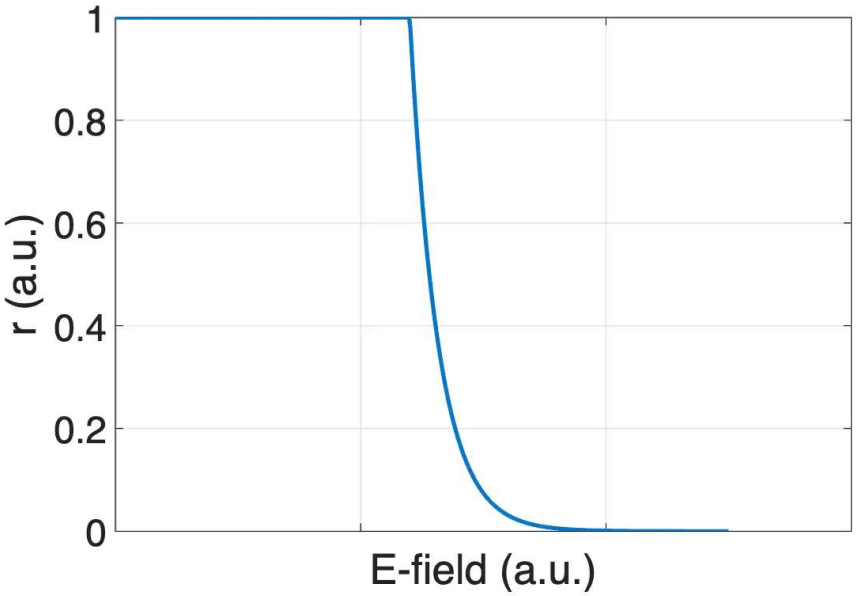
$$\frac{\partial \psi_1}{\partial r}\bigg|_{r=a} = \frac{\partial \psi_2}{\partial r}\bigg|_{r=a} \quad \rightarrow \quad A_1 \cdot \frac{u}{a} J_l'(u) = A_2 \cdot \frac{v}{a} K_l'(v)$$

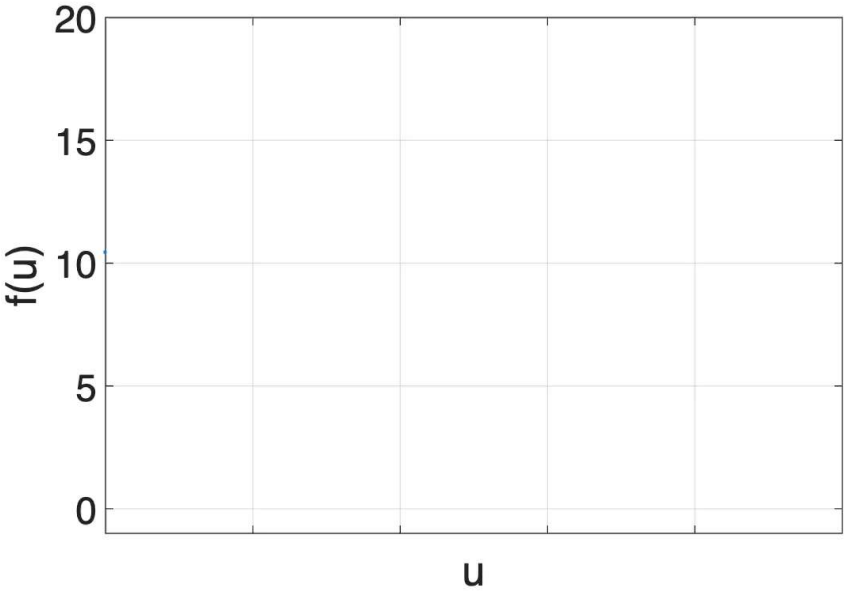
Linearly polarized (LP) waves

Linearly polarized (LP) waves



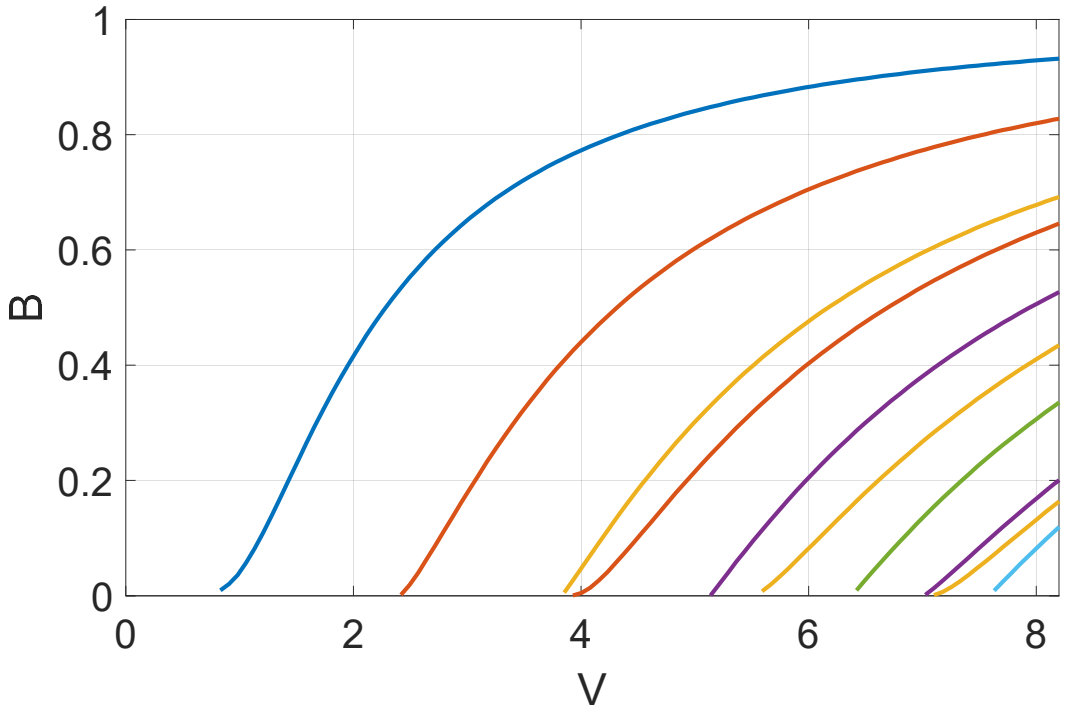
Linearly polarized (LP) waves





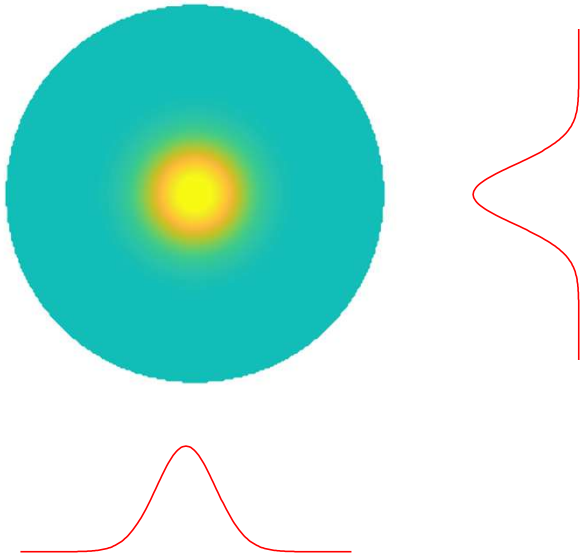
Linearly polarized (LP) waves

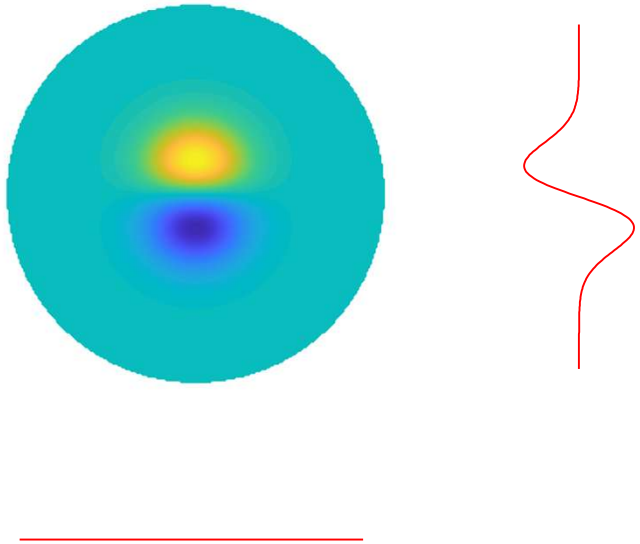
Linearly polarized (LP) waves



Mode Fields in Step Index Fibers

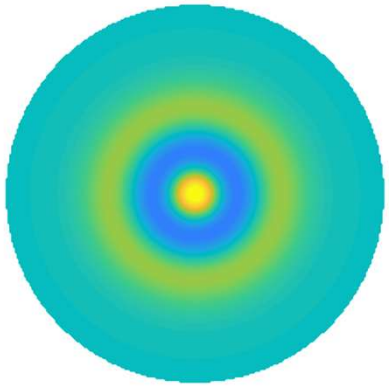
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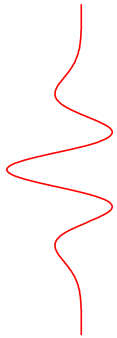


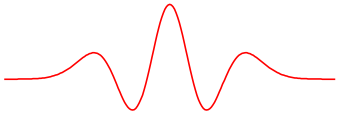


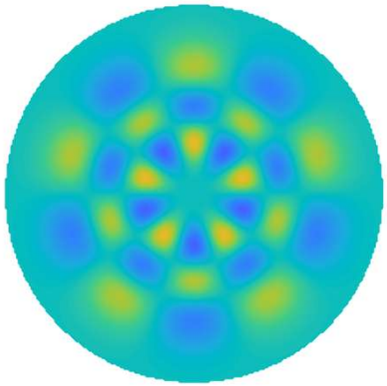
Mode Fields in Step Index Fibers

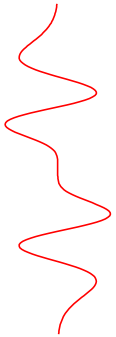
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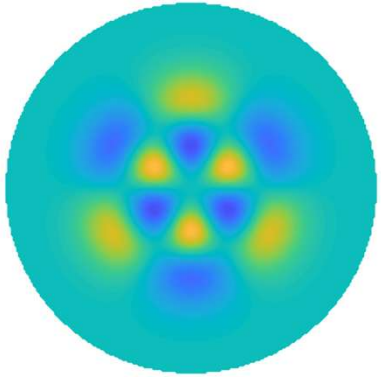
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Concept Test

Concept test

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Germany

Which mode is shown in the electric field profile of the fiber?



A LP23

B LP62

C LP32

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Single-mode fiber

Single-mode fiber

University of Stuttgart
Institute of electrical and optical communications

Thank you!



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